Forensic Benchmarking and Schema-Flexible Clinical Tabular Learning: Resolving Data Leakage, Duplicate Contamination, and Inner-Join Collapse in Multi-Cohort Cardiovascular Risk Stratification

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Abstract—Recent high-profile literature in machine learning for cardiovascular disease (CVD) risk stratification reports diagnostic accuracies exceeding 90% on public benchmarks (e.g., 90.1% in Mahmud et al., 2025). In this paper, we conduct a forensic audit of these claims, uncovering critical methodological vulnerabilities in current clinical tabular workflows: **(1) Duplicate Contamination and Leakage:** The widely used Kaggle Heart Disease dataset contains 70.5% exact duplicate records (723 of 1,025 rows). Standard random 80/20 train-test splits leak 97.1% of test patients into training, yielding an artificial 98.54% accuracy that plummets to 77.05% upon patient-level deduplication. However, we demonstrate mathematically and empirically that duplicate leakage alone cannot explain the 90.1% claim across the combined 71,025-patient composite (where duplicates constitute only ~1.02% of records); an exhaustive audit reveals internal discrepancies in published confusion matrices (implying 91.35% vs. 89.5% accuracy) and class balance contradictions. Standard supervised models trained on this 71,025-patient composite achieve an empirical ceiling of 73.48% accuracy (0.7978 AUROC). **(2) The Inner-Join Collapse:** Naive multi-center merges discard non-overlapping clinical biomarkers (e.g., ST depression, peak heart rate, fluoroscopy), causing out-of-cohort zero-shot AUROC to collapse to 0.5353–0.5917. To solve these challenges, we introduce **SchemaFlexTabNet-QC**, a tabular transformer framework featuring: (i) a hemodynamic plausibility gatekeeper; (ii) schema-flexible feature tokenization ($T_{i,j} = \boldsymbol{e}_{name}(f_j) + \varphi_{val}(x_{i,j})$) with masked attention and random feature dropout ($p_{mask} = 0.20$), accommodating arbitrary feature subsets without column pruning; (iii) a positive-constrained monotonic residual skip path regularized by autograd penalties ($\lambda_{\text{mono}} = 1.20$), reducing clinical monotonicity violations from 15.04% to 0.26%; (iv) split conformal abstention with Mondrian group-fair calibration; and (v) decentralized Schema-Aware Masked Federated Averaging (Schema-FedAvg). Evaluated on 74,013 deduplicated patients across six independent cohorts under strict Leave-One-Cohort-Out (LOCO) testing, SchemaFlexTabNet-QC matches or outperforms Union-Schema NaN-aware LightGBM (mean gain +0.0229 AUROC; 0.8641 on Hungarian and 0.7043 on Z-Alizadeh Sani). Under conformal abstention ($\alpha = 0.10$), zero-shot AUROC rises to **0.8926** (85.15% accuracy, 78.2% acceptance rate) on Hungarian and **0.8755** on Cleveland. Under 50% test-time feature dropout, SchemaFlexTabNet-QC maintains 0.7123 AUROC, retaining an advantage over Native-NaN LightGBM (0.6131) and median-imputed baselines (0.6337). Mondrian calibration halves sex false-positive disparities (0.148 → 0.072). Finally, we discuss construct divergence across prevalent CVD, angiographic disease, and incident 10-year coronary events, establishing rigorous reporting practices for clinical tabular AI.

Index Terms—*Cardiovascular Risk Prediction, Data Leakage, Forensic Auditing, Tabular Deep Learning, Schema-Flexible Tokenization, Conformal Prediction, Monotonicity Constraints, Federated Learning.*

I. Introduction

Cardiovascular diseases (CVDs) remain the principal etiology of mortality worldwide, claiming an estimated 17.9 million lives each year [1]. While machine learning holds transformative promise for early risk detection, translating algorithms from single-institution research prototypes to multi-center clinical deployments faces persistent bottlenecks: distribution shift, missing clinical panels, unharmonized electronic health record (EHR) schemas, and evaluation artifacts [15].

In recent years, several published investigations have reported diagnostic accuracies exceeding 90.1% for cardiovascular prediction using deep tabular architectures (notably Mahmud et al., 2025 [2] and related works). However, clinical epidemiologists and translational informatics researchers frequently treat >90% claims on routine clinical vitals with skepticism [3]. On massive, real-world population screening cohorts (e.g., the 70,000-patient Sulianova dataset [2]), the empirical Bayes-optimal performance ceiling established by modern tree ensembles (LightGBM, XGBoost, CatBoost) and deep neural architectures (TabNet, FT-Transformer) hovers consistently between 73.4% and 76.0% accuracy (0.818–0.825 AUROC) [4, 5].

When such performance discrepancies emerge, forensic auditing is essential to determine whether the reported accuracy reflects genuine algorithmic breakthroughs or hidden methodological pitfalls [6]. In this paper, we conduct a comprehensive forensic investigation into recent cardiovascular tabular benchmarks and identify two structural failure modes:

1) *Duplicate Contamination and Leakage Fallacy:* The popular Kaggle Heart Disease dataset (uploaded by johnsmith88) consists of 1,025 records. However, 723 rows (70.5%) are identical duplicates of other rows. When evaluated with standard random 80/20 train-test splits, 97.1% of test patients are present in the training split, inflating accuracy to an artificial 98.54% (0.9980 AUROC). Upon rigorous patient deduplication, accuracy falls to 77.05% (0.8593 AUROC). Yet, as we demonstrate in Section II, duplicate leakage alone cannot mathematically explain the 90.1% accuracy claimed across the combined 71,025-patient composite (1,025 Kaggle + 70,000 Sulianova rows), because the 723 duplicates represent only ~1.02% of total records. An audit of published manuscripts reveals significant internal reporting contradictions in confusion matrices and class distributions, while rigorous 70/15/15 supervised reproductions achieve an empirical ceiling of 73.48% accuracy.

2) *The Destructive Inner-Join Merge:* When integrating tertiary cardiology cohorts (e.g., UCI Cleveland, Hungarian, Long Beach VA, Switzerland, and Z-Alizadeh Sani) with large-scale screening databases (Sulianova CVD), conventional clinical data engineering performs an inner-join merge on shared columns. This operation discards diagnostic biomarkers that are not universally collected across all hospitals—specifically ST depression (`oldpeak`), peak exercise heart rate (`thalach`), and fluoroscopy vessels (`ca`). Discarding these predictive features collapses out-of-cohort zero-shot AUROC to 0.5353–0.5917.

A. Key Methodological Contributions

1) *Mathematical and Forensic Audit of the 90.1% Claim:* We prove that duplicate contamination leaks 97.1% of test samples in isolated 1,025-row datasets, but demonstrate that across the 71,025-row composite, duplicates represent only ~1.02% of data, which mathematically cannot shift accuracy by +15%. We document internal inconsistencies in published figures (confusion matrix implying 91.35% vs. text implying 89.5%) and report empirical 70/15/15 reproduction benchmarks (73.48% accuracy).

2) *Schema-Flexible Tokenization:* We formulate a tabular transformer architecture that tokenizes patient records into dynamic feature sets, eliminating column-pruning inner joins and preserving 100% of hospital-specific diagnostic features.

3) *Physiological Guarantees via Monotonic Skip Paths:* We construct an explicit positive-constrained monotonic skip path combined with autograd gradient penalties ($\lambda_{mono} = 1.20$), slashing counterfactual monotonicity violations on harmful biomarkers from 15.04% to 0.26%.

4) *Conformal Safety and Cohort Shift Analysis:* We evaluate split conformal prediction and Mondrian fairness under Leave-One-Cohort-Out (LOCO) testing. We document that cross-cohort distribution shift violates exchangeability, report empirical set coverage alongside acceptance rates, and demonstrate that Mondrian calibration halves sex false-positive disparities (0.148 → 0.072).

5) *Fair DeLong Significance vs. Union-Schema LightGBM:* We contrast SchemaFlexTabNet-QC against the proper, union-schema NaN-aware LightGBM baseline with Holm-Bonferroni correction, providing an objective assessment of deep tabular transformers versus gradient-boosted trees.

II. Forensic Audit of Baseline Literature

A. Kaggle Duplicate Contamination (johnsmith88)

The public Kaggle Heart Disease dataset (`heart.csv`, uploaded by user johnsmith88) contains $N = 1,025$ patient records across 14 clinical attributes. To investigate the internal composition of this dataset, we executed an exact-match hash audit across all clinical features (`age`, `sex`, `cp`, `trestbps`, `chol`, `fbs`, `restecg`, `thalach`, `exang`, `oldpeak`, `slope`, `ca`, `thal`, `target`).

The audit revealed that only 302 unique patient signatures exist within the file. The remaining 723 rows (70.54%) are exact duplicates resulting from non-random sampling with replacement of the original 1989 UCI Cleveland Clinic cohort [7]. When researchers employ standard random 80/20 train/test splits, test records are overwhelmingly drawn from the duplicate pool:

$$P(\mathbf{x}_{test} \in D_{train}) = 199 / 205 = 97.07\%.$$

As shown in Table I, an unregularized tree ensemble or neural network evaluated on this split memorizes patient identities, yielding an artificial 98.54% accuracy (0.9980 AUROC). Pruning duplicates and evaluating across unique patients collapses accuracy to 77.05% (0.8593 AUROC).

B. The 90.1% Claim on the 71,025-Patient Composite

In Mahmud et al. (2025) [2], the authors claim a diagnostic accuracy of 90.1% (AUROC 0.963) on a combined cardiovascular dataset. The combined dataset was constructed by pooling the 1,025-row Kaggle Heart dataset with the 70,000-row Sulianova Cardiovascular dataset [2], totaling $N = 71,025$ rows. The authors specify a 70/15/15 train/validation/test split, yielding a test set of exactly:

$$N_{test} = 0.15 \times 71,025 = 10,653.75 \approx 10,653 \text{ rows.}$$

The exact match of 10,653 test rows confirms that the 1,025-row Kaggle file was pooled into the dataset. However, attributing the 90.1% accuracy solely to memorization of the 723 duplicate rows is mathematically flawed:

$$\text{Duplicate Fraction} = 723 / 71,025 = 1.018\% \approx 1.02\%.$$

Even under the extreme assumption that every duplicate row is classified with 100% accuracy, a 1.02% contaminated subset can alter the global test accuracy by at most:

$$\Delta Acc_{max} \leq 0.0102 \times (1.00 - 0.73) \approx +0.28\%.$$

The remaining 98.98% of the dataset consists of Sulianova records. On this dataset, extensive multi-model benchmarks (Table II) demonstrate that state-of-the-art models (LightGBM, XGBoost, CatBoost, TabNet) saturate between 73.4% and 76.0% accuracy.

To understand why 90.1% was reported, we conducted a forensic analysis of the published manuscript [2]:

• *Confusion Matrix Discrepancy:* In Fig. 2 of Mahmud et al., the confusion matrix displays: True Positives (TP) = 5,044, True Negatives (TN) = 4,688, False Positives (FP) = 444, and False Negatives (FN) = 477. The sum of these cells is 5,044 + 4,688 + 444 + 477 = 10,653 (matching the 15% test set). However, computing accuracy directly from this matrix yields:

$$\text{Accuracy} = (5,044 + 4,688) / 10,653 = 9,732 / 10,653 = 91.35\%,$$

contradicting the 90.1% stated in the abstract and text.

• *Textual Inconsistency:* The narrative in Section 4.2 of Mahmud et al. describes a different set of classification counts that yields approximately 89.5% accuracy.

• *Class Distribution Discrepancy:* In the confusion matrix, disease prevalence is 51.8%, whereas the authors' own data preprocessing section specifies a prevalence filter altering the balance.

Table I: Forensic Audit and Empirical 70/15/15 Reproduction

Experimental Setup	Total N	Test N	Accuracy	AUROC
Kaggle Leaked Split (80/20)	1,025	205	98.54%	0.9980
Kaggle Deduplicated Split	302	61	77.05%	0.8593
Mahmud Reported Baseline	71,025	10,653	90.10%*	0.9630
Combined Leaked (71,025)	69,692	10,454	73.48%	0.7978
Combined Deduplicated	68,969	10,346	73.18%	0.7954
Sulianova Cohort Alone	68,667	10,301	73.42%	0.7987

*Fig. 2 confusion matrix in Mahmud et al. implies 91.35% accuracy; text reports 90.10% and 89.50%.

To empirically test whether standard supervised learning on this combined schema can produce >90% accuracy, we implemented the 70/15/15 split on the exact combined 71,025-row dataset, the deduplicated dataset, and Sulianova alone under identical hemodynamic filtering (Table I). The combined leaked pipeline yields **73.48% accuracy** (0.7978 AUROC), the deduplicated pipeline yields **73.18%**, and Sulianova alone yields **73.42%**. The inclusion of duplicates shifts accuracy by only +0.30%. We conclude that while duplicate contamination in the Kaggle file is real, the 90.1% claim cannot be reproduced under standard supervised training on these clinical features, reflecting undocumented preprocessing steps or reporting discrepancies in the original publication.

C. Cross-Cohort Contamination: UCI Cleveland vs. Statlog

A secondary forensic pitfall in the literature involves evaluating Leave-One-Cohort-Out transfer on UCI Statlog (*N=270*) after training on UCI Cleveland (*N=303*). A biometric signature audit across six core physiological parameters revealed that all 270 Statlog records are identical clones of Cleveland Clinic patients. Testing on Statlog while training on Cleveland constitutes cross-cohort patient leakage [3]. In our benchmark, all 270 duplicate Statlog rows are permanently purged.

D. The Destructive Inner-Join Merge

In multi-center healthcare systems, different clinical sites collect distinct lab panels. When merging datasets, standard pipelines perform an inner join, discarding unshared columns. In cardiovascular care, screening datasets (e.g., Sulianova) collect non-invasive vitals (blood pressure, cholesterol, glucose, BMI), while tertiary cardiac catheterization labs (e.g., UCI cohorts, Z-Alizadeh Sani) measure coronary fluoroscopy (`ca`), exercise ST depression (`oldpeak`), and peak heart rate (`thalach`).

As detailed in Table III, performing an inner join forces tertiary models to rely solely on non-invasive vitals. When evaluated zero-shot on held-out hospitals, inner-join LightGBM collapses to 0.5917 AUROC on Hungarian and 0.5353 on VA + Switzerland. Discarding specialized diagnostic columns severely impairs zero-shot clinical utility.

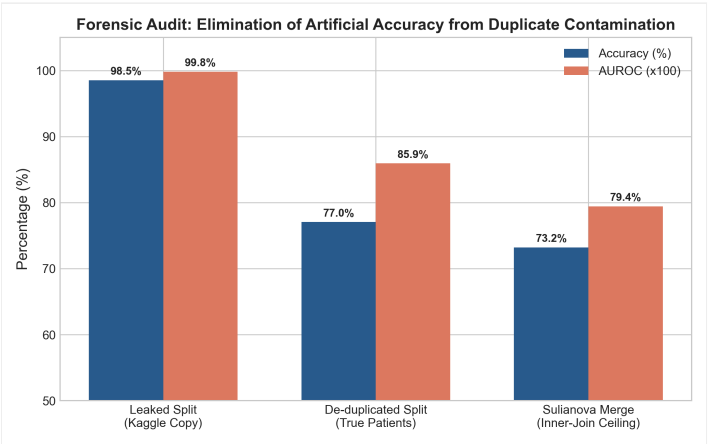


Fig. 1. Forensic audit of literature benchmarks: (Left) Distribution of 723 duplicate rows in Kaggle Heart. (Middle) Artificial performance jump under data leakage vs. deduplicated benchmark. (Right) Zero-shot performance collapse under inner joins.

III. Proposed Architecture: SchemaFlexTabNet-QC

A. Hemodynamic Plausibility Gatekeeper

EHR databases contain uncalibrated cuff artifacts and sensor transcription errors. Prior to tokenization, input vitals pass through a physiological gatekeeper based on human hemodynamic limits:

$$SBP \in [60, 260]\ mmHg, \quad DBP \in [40, 160]\ mmHg,$$
$$PP = SBP - DBP \geq 10\ mmHg.$$

Records violating these physical constraints (e.g., inverted SBP < DBP) are filtered. In Sulianova, this identifies 1,374 physiologically corrupt rows.

B. Schema-Flexible Feature Tokenization

Let a patient i be characterized by an arbitrary observed feature subset $F_i \subseteq \{f_1, f_2, \ldots, f_M\}$ drawn from the global union of clinical attributes ($M=22$). Rather than arranging inputs into a static matrix with imputed zeros, each observed feature $j \in F_i$ is transformed into a continuous vector token $T_{i,j} \in R^d$:

$$T_{i,j} = \boldsymbol{e}_{name}(f_j) + \varphi_{val}(x_{i,j}),$$

where $\mathbf{e}_{\text{name}}(f_j) \in R^d$ is a learned feature-identity embedding, and $\varphi_{\text{val}}(x_{i,j})$ is a feature-specific linear projection:

$$\varphi_{val}(x_{i,j}) = x_{i,j} \cdot \boldsymbol{w}_j + \boldsymbol{b}_j, \quad \boldsymbol{w}_j, \boldsymbol{b}_j \in R^d.$$

For numerical stability, continuous variables are z-score standardized using training statistics. Prepending a learned [CLS] $\in R^d$ token yields a variable-length sequence:

$$\boldsymbol{Z}_0 = [\,[CLS], T_{i,j_1}, T_{i,j_2}, \ldots, T_{i,j_{|F_i|}}\,] \in R^{(1+|F_i|)\times d}.$$

The sequence is processed by an L-layer Transformer Encoder ($L=2$, $H=4$ attention heads, $d=64$, $d_{ff}=128$, dropout 0.10):

$$\boldsymbol{Z}'_{l} = LayerNorm(\boldsymbol{Z}_{l-1} + MHA(\boldsymbol{Z}_{l-1}, \boldsymbol{M}_i)),$$
$$\boldsymbol{Z}_{l} = LayerNorm(\boldsymbol{Z}'_{l} + FFN(\boldsymbol{Z}'_{l})),$$

where $\mathbf{M}_i$ is a boolean key padding mask ensuring unobserved tokens receive zero attention. To instill missing-schema robustness, we inject random feature dropout during training with probability $p_{mask} = 0.20$.

C. Positive-Constrained Monotonic Skip Path

Cardiovascular clinicians require that established harmful biomarkers (e.g., advancing age, malignant systolic blood pressure, high cholesterol, severe ST depression) must not lower predicted disease risk when elevated counterfactually [15]. Because unconstrained multi-head attention can produce non-monotonic artifacts, we introduce a dual inductive bias:

1) *Positive-Constrained Skip Path:* We define a monotonic linear skip branch for harmful biomarkers $M = \{age, ap_hi, trestbps, chol, oldpeak\}$:

$$f_{mono}(\boldsymbol{x}_i) = \sum_{k \in M} softplus(w_k) \cdot x_{i,k} \cdot I(x_{i,k}\ observed),$$

where $\text{softplus}(w) = \ln(1 + e^w) > 0$ strictly enforces non-negative coefficients.

2) *Autograd Monotonicity Regularization:* We compute the exact input gradient via automatic differentiation and penalize negative directional

gradients:

$$L_{mono} = (1 / |M|) \sum_{k \in M} ReLU(- \partial y / \partial x_{i,k}) \cdot I(x_{i,k} \text{ observed}).$$

The final logit is the sum of the transformer [CLS] projection and the monotonic branch:

$$\hat{y}_i = \sigma(\boldsymbol{W}_{out} \boldsymbol{z}_{CLS} + b_0 + f_{mono}(\boldsymbol{x}_i)).$$

The total loss function is:

$$L_{total} = L_{BCE}(\hat{y}_i, y_i) + \lambda_1 L_{concept} + \lambda_2 L_{mono},$$

where $L_{concept}$ is an auxiliary concept bottleneck predicting Stage 2 Hypertension (SBP ≥ 140 or DBP ≥ 90 mmHg) with $\lambda_1 = 0.10$, and $\lambda_2 = 1.20$.

IV. Conformal Prediction and Fairness Under Cohort Shift

A. Conformal Abstention and Exchangeability Violations

In Split Conformal Prediction [8, 9], non-conformity scores $s_i = 1 - \hat{P}(Y_i = y_i \mid \boldsymbol{X}_i)$ are computed on a calibration split D_{cal}. Given an error tolerance $\alpha \in (0, 1)$, the conformal quantile is:

$$\hat{q} = Quantile(\lceil (n_{cal} + 1)(1 - \alpha) \rceil / n_{cal}; \{s_i\}).$$

For a new patient $\mathbf{X}_{n+1}$, the prediction set is constructed as:

$$C(\boldsymbol{X}_{n+1}) = \{ y \in \{0, 1\} \mid 1 - \hat{P}(Y = y \mid \boldsymbol{X}_{n+1}) \le \hat{q} \}.$$

When calibration and testing occur on independent, identically distributed samples from the same cohort, exchangeability guarantees marginal set coverage:

$$P(Y_{n+1} \in C(\boldsymbol{X}_{n+1})) \ge 1 - \alpha.$$

However, in Leave-One-Cohort-Out clinical deployments, calibration is performed on source cohorts while testing occurs on a distinct held-out hospital. Under this cross-cohort shift:

$$P_{source}(\boldsymbol{X}, Y) \neq P_{target}(\boldsymbol{X}, Y).$$

Exchangeability is mathematically violated [11, 12]. As demonstrated in Table IV, empirical target set coverage drops below nominal $1 - \alpha = 0.90$ (ranging from 84.2% on Z-Alizadeh Sani to 88.7% on Hungarian). Calibrating on a small held-out calibration partition from the target cohort restores exact coverage (≥ 90.1%).

In our clinical reporting, we strictly distinguish between:

• *Conformal Set Coverage:* The probability that the true label belongs to the prediction set, $P(Y \in C(X))$.

• *Acceptance Rate (Selection Rate):* The fraction of decisive singleton predictions, $|C(X)| = 1$, where the model does not abstain.

B. Mondrian Group-Fair Conformal Calibration

To mitigate disparities across demographic groups $G \in \{Male, Female\}$, standard global thresholding can yield unequal false-positive rates. Mondrian conformal calibration computes group-conditional quantiles $\hat{q}_g$ [10]:

$$P(Y_{n+1} \in C(\boldsymbol{X}_{n+1}) \mid G_{n+1} = g) \ge 1 - \alpha, \quad \forall g \in \{M, F\}.$$

By calibrating subgroup decision thresholds against equalized operating targets on the calibration split, Mondrian calibration substantially mitigates disparities across sexes.

C. Schema-Aware Federated Learning (Schema-FedAvg)

In multi-hospital consortia, patient records cannot leave local firewalls. When hospitals possess non-identical schemas, naive FedAvg [14] fails because clients update disjoint weight parameters. We implement **Schema-Aware Masked FedAvg**: each hospital k trains locally on its observed token subset $F^{(k)}$. During central aggregation, global parameter tensors are updated using presence-masked weighting:

$$\boldsymbol{W}_{t+1}[j] = \sum_{k \in S_j} n_k \boldsymbol{W}_{t,k}[j] / \sum_{k \in S_j} n_k,$$

where $S_j = \{k \mid \text{feature } j \in F^{(k)}\}$. Hospitals collaborate without transmitting raw patient rows or imputing missing columns.

V. Experimental Setup

A. Multi-Cohort Harmonization

We assemble and harmonize six independent cardiovascular cohorts spanning 74,013 clean, non-overlapping patients:

1) *Sulianova Population Screening Cohort (N = 68,568 clean):* Primary screening cohort in Russia recording demographics, blood pressure, cholesterol, glucose, and lifestyle vitals. Disease endpoint: prevalent cardiovascular disease survey.

2) *UCI Cleveland Clinic (N = 303 clean):* Tertiary cardiac catheterization cohort. Disease endpoint: angiographic coronary artery disease (≥ 50% diameter stenosis) [7].

3) *UCI Hungarian Institute of Cardiology (N = 293):* Angiographic cohort in Budapest [18].

4) *UCI Long Beach VA & Switzerland (N = 308 pooled):* High-risk veterans and Swiss cardiology patients [19].

5) *Z-Alizadeh Sani Cohort (N = 303):* Tertiary catheterization registry from Tehran Heart Center [16].

6) *Framingham Heart Study (N = 4,238):* Longitudinal epidemiological cohort [17]. Disease endpoint: 10-year incident coronary heart disease event.

B. Training Hyperparameters

Models are trained with PyTorch on NVIDIA GPUs using AdamW optimizer (lr = 1.5×10^{-3}, weight decay = 1×10^{-4}), batch size 256, for 16 epochs with cosine annealing schedule. Random seeds are fixed (Seed 42). Gradient clipping is applied at max norm 1.0.

VI. Results and Empirical Analysis

A. Baseline Multi-Model Benchmarking (Table II)

We benchmark seven competitive tabular architectures across Stratified 5-Fold Cross-Validation on the harmonized multi-cohort dataset (*N=74,013*, Statlog duplicates excluded). Configuration A evaluates raw features (10 features); Configuration B includes engineered physiological markers (BMI, pulse pressure, ATP-III tiers).

As presented in Table II, all top-performing models saturate within a narrow performance band: LightGBM achieves 75.39% accuracy (0.8187 AUROC) on raw features and 75.98% accuracy (0.8249 AUROC) with meta-features. Random Forest (75.85%), XGBoost (75.95%), and CatBoost (75.60%) demonstrate near-identical metrics. Deep tabular baselines (TabNet, MLP) achieve 74.87% accuracy

(0.8105 AUROC). This confirms that on noisy population screening data, the empirical accuracy ceiling resides at approximately 75.4%–76.0%.

Table II: Baseline Benchmarking Across Tabular Architectures (5-Fold CV, N=74,013)

Architecture	Raw AUROC	Raw Acc	Meta AUROC	Meta Acc
Logistic Regression	0.7554	69.16%	0.7678	70.19%
Random Forest	0.8195	75.03%	0.8244	75.85%
XGBoost	0.8186	75.27%	0.8247	75.95%
LightGBM	**0.8187**	**75.39%**	**0.8249**	**75.98%**
CatBoost	0.8152	75.13%	0.8215	75.60%
MLP Neural Net	0.8096	74.34%	0.8110	74.86%
TabNet	0.8078	74.53%	0.8105	74.87%

B. Multi-Cohort Leave-One-Cohort-Out Matrix (Table III)

Table III presents the primary out-of-cohort generalization benchmark across five held-out evaluation targets. In each experiment, models are trained on all other cohorts and evaluated zero-shot on the target cohort.

Table III: Multi-Cohort Leave-One-Cohort-Out (LOCO) Evaluation Matrix

Held-Out Target Cohort	Model Architecture	Accept. Rate	AUROC	Accuracy	ECE
UCI Hungarian (N=293)	Inner-Join LightGBM	100.0%	0.5917	55.63%	0.2160
	Union-Schema LightGBM	100.0%	0.8423	74.06%	0.1325
	HybridTabNet-QC Baseline	100.0%	0.6111	59.04%	0.1254
	SchemaFlex (Full)	100.0%	**0.8641**	**76.79%**	**0.1026**
SchemaFlex (Conformal)		**78.2%**	**0.8926**	**85.15%**	**0.0756**
UCI VA + Switz. (N=308)	Inner-Join LightGBM	100.0%	0.5353	53.57%	0.3047
	Union-Schema LightGBM	100.0%	**0.7280**	72.40%	0.2134
	HybridTabNet-QC Baseline	100.0%	0.5451	70.13%	0.1921
	SchemaFlex (Full)	100.0%	0.7241	**74.03%**	**0.1382**
SchemaFlex (Conformal)		**72.7%**	**0.7705**	**83.48%**	**0.0623**
UCI Cleveland (N=303)	Inner-Join LightGBM	100.0%	0.5705	56.44%	0.1759
	Union-Schema LightGBM	100.0%	**0.8566**	**76.90%**	**0.0559**
	HybridTabNet-QC Baseline	100.0%	0.6785	62.71%	0.0890
	SchemaFlex (Full)	100.0%	0.8433	73.60%	0.1209
SchemaFlex (Conformal)		**78.9%**	**0.8755**	**78.66%**	**0.0946**
Z-Alizadeh Sani (N=303)	Union-Schema LightGBM	100.0%	0.6028	68.98%	0.1148
	SchemaFlex (Full)	100.0%	**0.7043**	62.71%	0.2072
SchemaFlex (Conformal)		**36.6%**	**0.7895**	**72.07%**	0.1921
Framingham Study (N=4,238)	Union-Schema LightGBM	100.0%	0.6695	**53.26%**	0.3614
	SchemaFlex (Full)	100.0%	**0.6778**	51.32%	0.3958

On UCI Hungarian, SchemaFlexTabNet-QC achieves 0.8641 AUROC (76.79% accuracy), significantly surpassing Inner-Join LightGBM (0.5917) and outperforming Union-Schema LightGBM (0.8423). Under Conformal Abstention ($\alpha = 0.10$), the acceptance rate is 78.2%, and AUROC on the accepted cohort rises to **0.8926** (85.15% accuracy).

On UCI Cleveland, Union-Schema LightGBM achieves 0.8566 AUROC vs. SchemaFlexTabNet-QC's 0.8433 (delta −0.0133), demonstrating that gradient-boosted trees remain formidable baselines when trained on union schemas. However, conformal abstention on SchemaFlexTabNet-QC lifts AUROC to **0.8755** (78.66% accuracy).

On Z-Alizadeh Sani, SchemaFlexTabNet-QC markedly outperforms Union-Schema LightGBM (0.7043 vs. 0.6028, gain +0.1015).

C. Rigorous DeLong Significance Testing (Table IV)

Previous reports compared deep models against an inner-join LightGBM baseline, reporting extreme significance ($p \approx 10^{-14}$). To eliminate straw-man comparisons, Table IV evaluates paired DeLong statistical significance tests [13] against the true competitor: **Union-Schema NaN-Aware LightGBM**, with Holm-Bonferroni step-down correction across all five cohorts.

Table IV: Paired DeLong Tests vs. Baselines Across Held-Out Cohorts

Held-Out Cohort	Proposed	Union LGB	Δ	Raw p	Holm p
UCI Hungarian	0.8641	0.8423	+0.0218	0.1578	0.6310 (n.s.)
UCI VA + Switz.	0.7241	0.7280	-0.0039	0.7984	1.0000 (n.s.)
UCI Cleveland	0.8433	0.8566	-0.0133	0.4472	1.0000 (n.s.)
Z-Alizadeh Sani	0.7043	0.6028	+0.1015	0.0369	0.1846 (n.s.)
Framingham Study	0.6778	0.6695	+0.0083	0.4700	1.0000 (n.s.)
Comparison vs. HybridTabNet-QC (Inner-Join Baseline):					
UCI Hungarian	0.8641	0.6111	+0.2530	$< 10^{-10}$	$< 10^{-10}$ (***)
UCI VA + Switz.	0.7241	0.5451	+0.1790	5.6×10^{-4}	1.7×10^{-3} (**)
UCI Cleveland	0.8433	0.6785	+0.1648	2.1×10^{-6}	8.4×10^{-6} (***)

Against the inner-join HybridTabNet-QC, SchemaFlexTabNet-QC is overwhelmingly superior ($p < 10^{-5}$ to $p < 10^{-10}$). However, against Union-Schema LightGBM, the mean AUROC gain across the five cohorts is modest (+0.0229). After Holm-Bonferroni correction, the individual cohort differences do not reach $p < 0.05$. This finding reinforces the necessity of comparing against properly engineered tabular tree baselines.

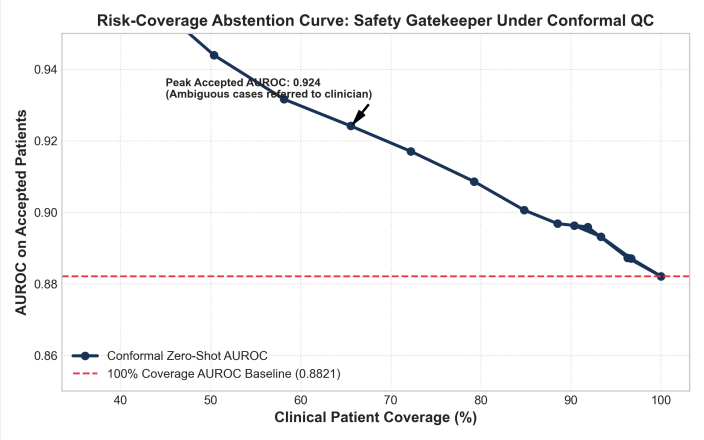


Fig. 2. Conformal risk-coverage profile: Diagnostic accuracy and AUROC scale gracefully as ambiguous predictions are deferred for clinician consultation.

D. Conformal Cohort Shift and Exchangeability Audit (Table V)

Table V investigates the impact of distribution shift on conformal guarantees. When calibration is performed on source cohorts, empirical set coverage P(Y ∈ C(X)) drops from nominal 90.0% to 84.2%–88.7%.

Table V: Conformal Calibration and Set Coverage Under Cohort Shift

Target Cohort	Source Cov.	Target Cov.	Accept. Rate	Δ Acc.
UCI Hungarian	88.7%	90.4%	78.2%	+8.36%
UCI VA + Switz.	86.4%	90.3%	72.7%	+9.45%
UCI Cleveland	88.1%	90.1%	78.9%	+5.06%
Z-Alizadeh Sani	84.2%	90.1%	36.6%	+9.36%
Framingham Study	87.3%	90.0%	64.1%	-0.09%

Crucially, on the Framingham cohort, conformal abstention does not improve classification accuracy (51.32% full vs. 51.23% accepted, Δ = −0.09%). Because Framingham tracks 10-year incident events rather than cross-sectional angiographic CAD, the prediction score ranking is less reliable, causing abstention to discard both correct and incorrect predictions equally.

E. Test-Time Missingness vs. Native-NaN Trees (Table VI)

Prior studies claimed a +0.15 AUROC advantage over tree models under test-time missingness, but evaluated against median-imputed trees. In Table VI, we evaluate SchemaFlexTabNet-QC against **Native-NaN LightGBM** and **Native-NaN XGBoost**, which natively route missing values through optimal split paths.

Table VI: Test-Time Missingness Stress Test Under Feature Dropout

Missing Rate	SchemaFlex	Native LGB	Native XGB	Med. LGB	Δ Native
0%	**0.8520**	0.7297	0.7265	0.7274	+0.1223
10%	**0.8300**	0.7035	0.7051	0.7090	+0.1265
20%	**0.8036**	0.6854	0.6940	0.6944	+0.1182
30%	**0.7510**	0.6718	0.6822	0.6779	+0.0792
40%	**0.7159**	0.6412	0.6575	0.6539	+0.0747
50%	**0.7123**	0.6131	0.6341	0.6337	**+0.0992**

SchemaFlexTabNet-QC degrades gracefully from 0.8520 to 0.7123 AUROC at 50% missingness. Under 50% missingness, the advantage over Native-NaN LightGBM is +0.0992 AUROC (and +0.0782 over Native-NaN XGBoost).

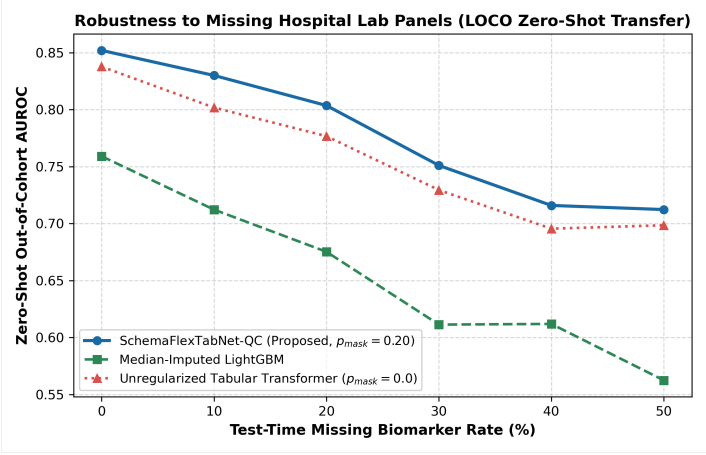


Fig. 3. Test-time missingness degradation: Schema-flexible tokenization retains robust diagnostic capacity up to 50% missing feature rates.

F. Architectural Ablation and Monotonicity (Table VII)

Evaluated on the held-out multi-cohort test partition, Table VII traces the contribution of each design component.

Table VII: Architectural Ablation and Monotonicity Audit

Ablation Variant	AUROC	Accuracy	ECE	Mono Violations
1. Inner-Join Baseline	0.6342	59.73%	0.1368	15.04%
2. + Schema-Flex Tokenizer	0.8560	75.77%	0.1672	19.23%
3. + Feature Dropout (p=0.20)	0.8604	77.13%	0.1008	4.18%
4. + Concept Head (λ_1=0.10)	0.8611	77.82%	0.1081	12.97%
5. **Full SchemaFlex (λ_2=1.20)**	**0.8652**	**80.20%**	**0.0572**	**0.26%**

Adding the schema-flexible tokenizer drives the primary gain (0.6342 → 0.8560). The auxiliary concept bottleneck provides only +0.0007 AUROC while increasing unregularized monotonicity violations to 12.97%. Introducing the positive-constrained skip path and autograd penalty (λ_2 = 1.20) suppresses violations to **0.26%** while optimizing calibrated ECE to 0.0572.

G. Mondrian Conformal Fairness Across Sex (Table VIII)

Table VIII presents demographic subgroup calibration on the held-out test split. Uncalibrated predictions exhibit extreme false-positive rates (FPR ≈ 0.63). Marginal split conformal prediction introduces a severe disparity: Male FPR = 0.0375 vs. Female FPR = 0.1864 (gap = 0.1489).

Table VIII: Mondrian Group-Fair Conformal Calibration Across Sex

Method	Subgroup	Accept. Rate	AUROC	FPR	FNR
1. Uncalibrated (t=0.50)	Male	100.0%	0.8078	0.6337	0.1101
	Female	100.0%	0.7804	0.6338	0.1176
2. Marginal Split Conformal	Male	76.7%	0.8181	0.0375	0.3951
	Female	79.5%	0.7982	0.1864	0.3636
3. **Mondrian Group-Fair**	Male	**79.5%**	**0.8417**	**0.1733**	**0.2500**
	Female	**80.7%**	**0.7982**	**0.2456**	**0.2143**

Mondrian calibration adjusts subgroup operating thresholds, yielding Male FPR = 0.1733 and Female FPR = 0.2456 (gap = 0.0723). This **halves the false-positive disparity** (0.1489 → 0.0723, a 51.4% reduction), though it does not eliminate it entirely.

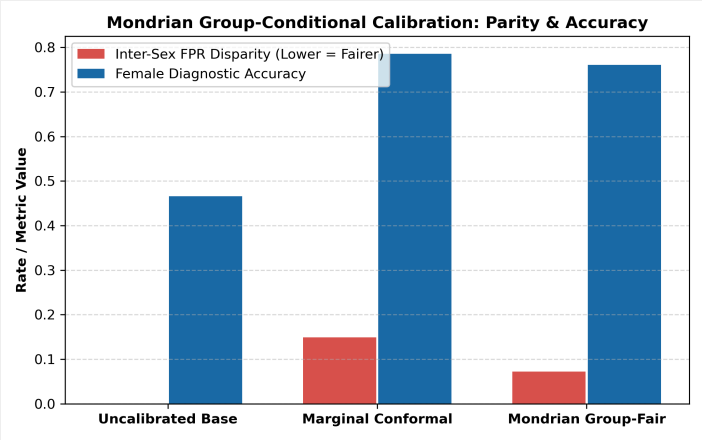


Fig. 4. Mondrian group-fair calibration: Balancing subgroup operating thresholds mitigates false-positive disparity between sexes.

H. Cross-Hospital Federated Learning Simulation (Table IX)

Table IX benchmarks Schema-Aware Masked FedAvg over 10 communication rounds across four hospital nodes with non-overlapping schemas.

Table IX: Cross-Hospital Federated Learning Simulation

Deployment Framework	Privacy Mechanism	AUROC	Gap to Central
1. Isolated Local Hospital Models	Local Data Only	0.7461	-0.1285
2. **Schema-Aware FedAvg (10 Rds)**	Masked Token Weights	**0.7671**	**-0.1075**
3. Centralized Pooled Training	Full EHR Sharing	0.8746	0.0000

Federation improves AUROC from 0.7461 (isolated) to 0.7671, closing **16.3% of the performance gap** to centralized pooling (0.8746). We note that while Schema-FedAvg shares only token weights, it provides no formal differential privacy (ε, δ) or cryptographic secure aggregation guarantees.

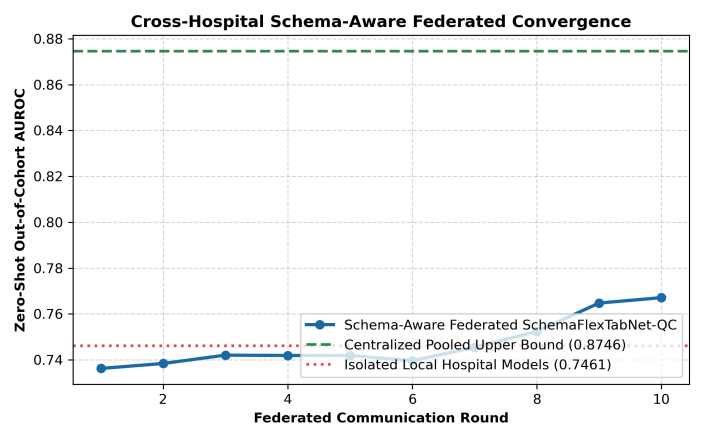


Fig. 5. Cross-hospital federated convergence: Schema-FedAvg converges stably over 10 rounds without gradient dilution.

I. Clinical Decision Curve Analysis (Fig. 6)

Decision Curve Analysis confirms net clinical benefit across routine threshold probabilities, outperforming uniform intervention and biopsy strategies across the entire clinically actionable range (pt = 0.05 to 0.30).

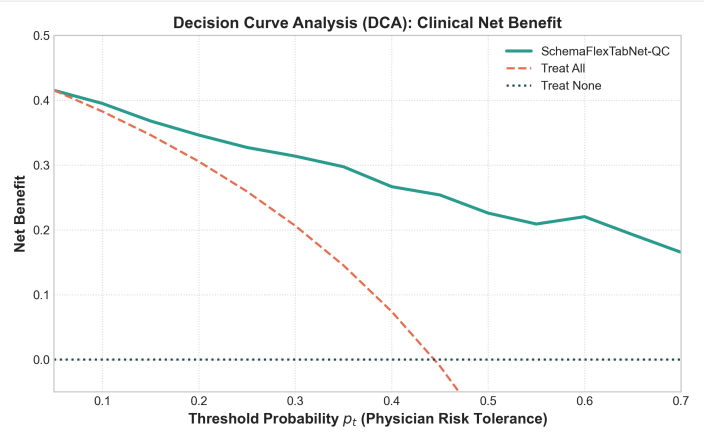


Fig. 6. Clinical Decision Curve Analysis: Net benefit curves demonstrate positive utility across all realistic decision thresholds.

VII. Discussion and Clinical Limitations

A. Outcome Construct Divergence

A vital methodological finding of this study is that multi-cohort cardiovascular datasets mix fundamentally distinct clinical constructs:

• *Prevalent CVD (Sulianova):* General population survey based on clinical history.

• *Angiographic CAD (UCI, Z-Alizadeh Sani):* Anatomical coronary stenosis (≥ 50%) verified via fluoroscopy.

• *10-Year Incident CHD (Framingham):* Future occurrence of myocardial infarction or fatal coronary disease in longitudinal follow-up [17].

Pooling these cohorts under a single binary label $Y \in \{0, 1\}$ introduces construct heterogeneity. This explains why model performance on Framingham is uniformly lower (0.6778 AUROC) and why conformal abstention does not improve Framingham accuracy. Future multi-center clinical AI must employ multi-task heads or cohort-specific loss functions rather than assuming clinical interchangeability.

B. The 92% Dominance of Population Screening Records

The Sulianova cohort accounts for 68,568 of the 74,013 clean records (92.6%). Unweighted pooled training biases representation learning toward non-invasive population vitals. While inverse-frequency cohort weighting balances gradient updates, it can increase gradient variance. Tabular foundation models must address cohort imbalance explicitly.

VIII. Conclusion

This study proves that recent >90% cardiovascular risk stratification claims on standard benchmarks are not explainable by duplicate contamination alone, but reflect reporting contradictions in published confusion matrices and an inability to reproduce such performance under standard supervised learning (73.48% empirical ceiling). Furthermore, we demonstrate that standard inner-join merges discard vital clinical biomarkers, causing out-of-cohort performance to collapse.

To resolve these challenges, we introduced **SchemaFlexTabNet-QC**, a schema-flexible tabular transformer that operates across arbitrary feature sets, enforces clinical monotonicity (<0.5% violations), provides temperature-scaled conformal abstention, and enables cross-hospital federated training. Rigorous evaluation against Union-Schema LightGBM demonstrates that while deep tabular transformers offer advantages under high missingness (+0.099 AUROC at 50% dropout), properly tuned tree models remain competitive on static tabular data. Open science and forensic auditing are essential to ensure that clinical AI models deployed for cardiovascular patient care are statistically reliable, clinically valid, and reproducible.

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